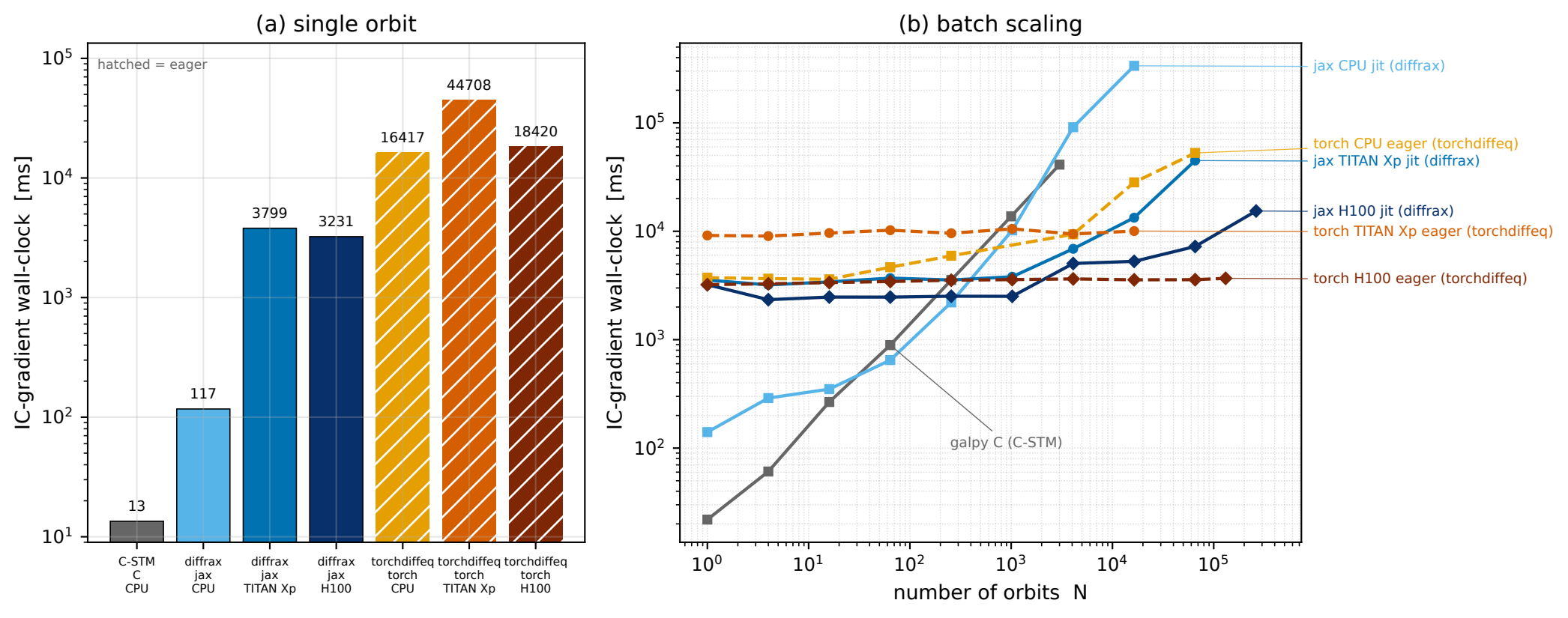
(a) single orbit
hatched = eager
IC-gradient wall-clock [ms]
10^5
10^4
10^3
10^2
10^1
13
117
3799
3231
16417
44708
18420
C-STM C CPU
diffrax jax CPU
diffrax jax TITAN Xp
diffrax jax H100
torchdiffeq torch CPU
torchdiffeq torch TITAN Xp
torchdiffeq torch H100
(b) batch scaling
IC-gradient wall-clock [ms]
10^5
10^4
10^3
10^2
10^0
10^1
10^2
10^3
10^4
10^5
number of orbits N
jax CPU jit (diffrax)
torch CPU eager (torchdiffeq)
jax TITAN Xp jit (diffrax)
jax H100 jit (diffrax)
torch TITAN Xp eager (torchdiffeq)
torch H100 eager (torchdiffeq)
galpy C (C-STM)